1 Discourse of 19th february: the parameter analysis

There given a sample set of interventions

$$V_{\mathrm{do}} = \{\mathrm{do}(V = v_0), \mathrm{do}(V = v_1), \dots\},$$

where v_i (is a) set of all nodes.

1. Assume we have samples of V_{do} (with «a» distribution) and the variables for any $V \in v$
$$S_1^v, \dots S_n^v.$$

2. For any $y \in V$, compute $\mathrm{Anc}(y)$ by:
$$\mathrm{Anc}(y) = \{v, y \not\perp\!\!\!\perp V_{\mathrm{do}}\}.$$

3. For each, $\forall Y \in V, \forall x \in \mathrm{Anc}(y)$:
   (a) estimate $w_i^{yx} = \arg\max_w P(y \mid X_{\mathrm{do}}; w)$, $1 \leq i \leq n$ and obtain $\hat{w}_1^{yx}, \dots, \hat{w}_n^{yx}$, where $P(y \mid X_{\mathrm{do}}; w)$ in the probability of Y from $\mathrm{Anc}(Y)$ based on all samples, assessed with X;
   (b) compute $\hat{\boldsymbol{\mu}}^{yx}$ and $\hat{\boldsymbol{\Sigma}}^{yx}$ from $\hat{w}_1^{yx}, \dots, \hat{w}_n^{yx}$;
   (c) compute $D_{\mathrm{KL}}\Big(p(w; \hat{\boldsymbol{\mu}}, b\hat{S}ig) \,\Big\|\, p(w; \hat{\boldsymbol{\mu}}^{xy}, \hat{\boldsymbol{\Sigma}}^{xy})\Big)$;
   (d) if $D_{\mathrm{KL}}(\,\|\,) \geq 0$, $x \in p_a(Y)$.

The first variant

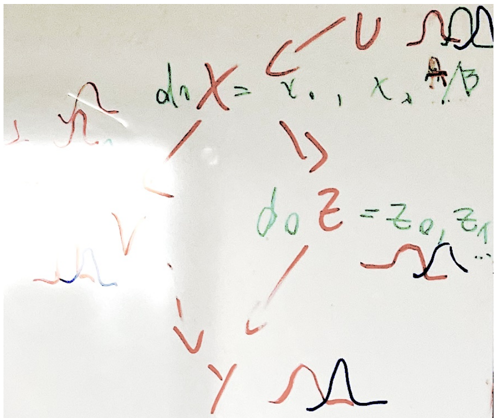
$$X_{\mathrm{do}}\{\mathrm{do}(X = x_0), \mathrm{do}(X = x_1), \dots\};$$

V_i set of all nodes
(a) If $X_{\mathrm{do}} \not\perp\!\!\!\perp Y$, then $X \in \mathrm{Anc}(Y)$

- If there exists $\mathrm{do}(Z)$, where $Z = V \setminus X$, s.t. $X \perp\!\!\!\perp Y \mid \mathrm{do}(Z)$, then $X \notin Pa(Y)$.
- Otherwise, $X \in \mathrm{Pa}(Y) \to$ impossible in practice

(b) With (a) we can compute the ancestors of all variables , e.q.

$$\begin{array}{rl}
\mathrm{Anc}(Y) = \{V, Z, X, V\}, \mathrm{Anc}(X) = \{V\} & => \mathrm{Pa}(X) = \{V\} \\
\mathrm{Anc}(V) = \{X, V\}, \mathrm{Anc}(V) = \{\emptyset\} & => \mathrm{Pa}(V) = \{\emptyset\} \\
\mathrm{Anc}(Z) = \{X, V\}, & => ?Z \perp\!\!\!\perp X \mid V
\end{array}$$


U
do X =
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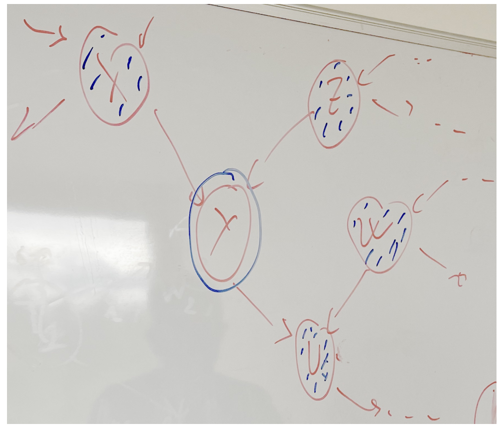
Y



Direct and indirect causes in the time series

IA pour la Science

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Introduction

The goal of this document is to describe the target function of the causal graph.

We treat the algorithm PCGCE in the paper *Extended summary graphs (ESG)* belongs to class of the exhaustive search algorithms. So to an optimal causal graph we make no difference between

1) exhaustive printing or combinatorial algorithms,

2) gradient-based algorithms including algorithms with relaxations,

3) zero-gradient algorithms.

The only thing we expect its result is a causal graph, optimal according to this criterion.

The model structure includes:

1) the time is continuous $\frac{dx}{dt} = f(x,t)$ (basic: discrete with $\Delta t = 1$),

2) the time series X, Y, Z has a spacial structure $\frac{dx}{dx} = g(x,s)$ (basic: scalar-valued),

3) there is no limitations of the time as in the ESG,

4) the model is linear in the latent space (basic: linear in the observable space),

5) the latent space represent a spectrum of the observable domain.

The graph structure:

1) the graph is a bipartite graph between the observable and the latent,

2) the direct graph is isomorphic to the reverse graph,

3) the causal graph is in the hypergraph that groups the variables or the time.

The time structure for the convolutional model $\int_{\tau=0}^{T} f(\tau)h(t-\tau)d\tau$

The causal graph $G = G(t)$ is a time series. We suppose either:

1) it is stationary fot all t,

2) it is stationary for $t \geq T$ for with the period T,

3) it is non-stationary but with rejection to discover its structure.

2 Appendix. Natural Gradient Descent

The natural gradient descent is a method to optimize the model parameters θ by estimating the distribution

$$p(\theta) \propto \exp(-\mathcal{L}(\theta))$$

where $\mathcal{L}(\theta)$ is the loss function of the model. The natural gradient descent is defined as

$$\theta_{t+1} = \theta_t - \eta F^{-1}$$

where F is the Fisher information matrix of the model parameters θ.

Fisher Metric and Natural Gradient

When working with probabilistic models, we can use a metric that is more natural for probability distributions: the Kullback–Leibler (KL) divergence. Suppose we have a likelihood model

$$p(x \mid \theta),$$

where x denotes fixed data and θ is a parameter vector.

Let θ_t denote the current parameter values, and suppose we wish to find updated parameters θ_{t+1}. Under the natural gradient framework, this is done by solving the following optimization problem:

$$\theta_{t+1} = \arg\min_{\theta} \left[f(\theta_t) + \nabla f(\theta_t)^\top (\theta - \theta_t) + D_{\mathrm{KL}}(p(x \mid \theta) \parallel p(x \mid \theta_t)) \right].$$

In general, this minimization problem is intractable. However, the KL divergence can be approximated locally using a second-order Taylor expansion. This expansion yields the Fisher information matrix F, which captures the local curvature of the statistical manifold. In particular, around θ_t, we have

$$D_{\mathrm{KL}}(p(x \mid \theta) \parallel p(x \mid \theta_t)) \approx \frac{1}{2}(\theta - \theta_t)^\top F (\theta - \theta_t),$$

where the Fisher information matrix is defined as

$$F = \mathbb{E}_{x \sim p(x|\theta)} \left[\left(\nabla_\theta \log p(x \mid \theta)\right) \left(\nabla_\theta \log p(x \mid \theta)\right)^\top \right].$$

The Fisher information matrix therefore encodes the curvature of the likelihood-based loss function in parameter space. Using this approximation, the resulting natural gradient update rule becomes

$$\theta_{t+1} = \theta_t - \gamma F^{-1} \nabla f(\theta_t),$$

where $\gamma > 0$ is the learning rate.

Natural Gradient from an Information-Geometric Perspective

Consider a parametric family of probability distributions

$$\mathcal{M} = \{p(x \mid \theta) \mid \theta \in \Theta \subset \mathbb{R}^d\},$$

which forms a smooth statistical manifold $\mathcal{M}$. Each parameter vector θ corresponds to a point on this manifold, and the coordinate system θ is generally non-Euclidean.

The natural notion of distance between nearby probability distributions on $\mathcal{M}$ is given by the Kullback–Leibler divergence. For two infinitesimally close distributions $p(x \mid \theta)$ and $p(x \mid \theta + d\theta)$, the KL divergence admits the second-order expansion

$$D_{\mathrm{KL}}(p(x \mid \theta + d\theta) \parallel p(x \mid \theta)) = \frac{1}{2} d\theta^\top G(\theta) d\theta + o(\|d\theta\|^2),$$

where

$$G(\theta) = \mathbb{E}_{x \sim p(x|\theta)} \left[\left(\nabla_\theta \log p(x \mid \theta)\right) \left(\nabla_\theta \log p(x \mid \theta)\right)^\top \right]$$

is the **Fisher information metric tensor** on $\mathcal{M}$.

Let $f(\theta)$ be a smooth objective function defined on the manifold (e.g., the negative log-likelihood or empirical risk). Given the current point $\theta_t \in \mathcal{M}$, we seek a new point θ_{t+1} by minimizing a first-order approximation of f subject to a constraint on the geodesic distance induced by the Fisher metric. This leads to the optimization problem

$$\theta_{t+1} = \arg\min_\theta \left\{ f(\theta_t) + \nabla f(\theta_t)^\top (\theta - \theta_t) \;\middle|\; D_{\mathrm{KL}}(p(x \mid \theta) \parallel p(x \mid \theta_t)) \le \epsilon \right\}.$$

Using the local quadratic approximation of the KL divergence, this constrained problem reduces to minimizing the linearized objective under a Riemannian norm constraint. The solution is given by a step in the direction of the **natural gradient**, defined as

$$\widetilde{\nabla} f(\theta) = G(\theta)^{-1} \nabla f(\theta).$$

Accordingly, the natural gradient descent update takes the form

$$\theta_{t+1} = \theta_t - \gamma \widetilde{\nabla} f(\theta_t) = \theta_t - \gamma G(\theta_t)^{-1} \nabla f(\theta_t),$$

where $\gamma > 0$ is the step size.

From an information-geometric standpoint, this update corresponds to moving along the direction of steepest descent with respect to the Fisher–Rao Riemannian metric, ensuring invariance under smooth reparameterizations of the statistical manifold.